

Noncorrelated Automatic Sequences: What They Can and Cannot Do for Monte Carlo Integration

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Victoria Portnaya
Applied Mathematics Program, Department of Mathematics
Kyiv School of Economics
Mentor: Jakub Konieczny

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Abstract

This project asks whether a deterministic binary sequence can be useful for Monte Carlo integration when it has no lasting correlation with any fixed shift of itself. The sequences studied here are built by counting chosen binary patterns. For a finite set of words A , the value at n is $a_{A,k}(n) = \omega_k^{\#(A,n)}$. A sequence is called noncorrelated (NC) when its limiting autocorrelation is zero at every nonzero lag. Informally, an NC sequence and any fixed shift of it eventually agree and disagree in a balanced way. This resembles one feature of independent random signs, but it does not immediately guarantee good multidimensional sample points.

The first part of the report develops the mathematics needed to make this condition precise. A pattern-counting sequence is 2-automatic because an automaton only has to remember a short suffix of the binary input and the pattern count modulo k . Autocorrelation is then expressed as a Fourier coefficient of a simple residue distribution. Herglotz's theorem gives a second interpretation: for sequences of absolute value one, the NC condition is exactly the statement that the autocorrelation spectrum is flat. The Rudin–Shapiro sequence is treated in detail. Its NC property is proved by splitting the correlation sums into even and odd indices; this calculation also shows why one extra, parity-weighted correlation must be tracked.

The second part checks these ideas computationally. Two censuses contain 846 and 2047 pattern sets, respectively. They are tested for moduli from 2 to 8 and for lags up to 64. Exact finite-state recursions are compared with independently generated prefixes, Fejér spectra, kernel calculations, and 1400 experiments on transformations of the sequences. These checks do not replace a proof, but they help detect coding or interpretation errors. In the named binary catalog, all ten examples that pass the operational NC test follow an empirical $1/N$ law for the overlap-prefix statistic used in the computation, for $N = 2^{10}, \dots, 2^{18}$. The sixteen non-NC controls do not decay to zero on this range; instead, they settle near nonzero values.

Finally, two ways of using the same sequences are tested for numerical integration. Directly packing their bits into coordinates works poorly. In the exhaustive European-call experiment with $N = 8192$, the median absolute error is 0.2362. A much better approach is to keep a Sobol point set and use the sequence only to choose a digital mask. The median error then falls to 7.55×10^{-4} . A controlled comparison uses the same Sobol points, sample sizes, number of masks, and reference values for four NC families and for uniform random shifts. NC shifts win 18 of the 32 final-size comparisons, but their geometric-mean RMSE ratio relative to random shifts is 0.9894. This is effectively a tie.

The main lesson is therefore about the roles played by the two ingredients. NC is a useful and exactly computable property of a one-dimensional sequence, but it is not a theorem

about low discrepancy and it says little about higher-order blocks of bits. In the successful hybrid method, the Sobol net provides the multidimensional balance, while the NC sequence provides a fixed, reproducible mask. Keeping these roles separate makes it clear which claims are proved, which are observed in finite experiments, and which are not supported by the results.

Keywords: automatic sequences; Rudin–Shapiro; autocorrelation; spectral measure; Monte Carlo; quasi-Monte Carlo; Sobol nets; digital shifts; discrepancy; reproducible computation.

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1 Problem setup and goals

1.1 The integration problem

Let $f : [0, 1]^d \rightarrow \mathbb{R}$ be an integrable function. The goal is to approximate its integral by averaging the values of f at sample points:

$$I(f) = \int_{[0,1]^d} f(\mathbf{u}) \, d\mathbf{u}, \quad Q_N(f) = \frac{1}{N} \sum_{n=0}^{N-1} f(\mathbf{u}_n). \quad (1)$$

In ordinary Monte Carlo, the points are independent and uniformly distributed. If the integrand has finite variance, this gives root-mean-square error of order $N^{-1/2}$. Quasi-Monte Carlo (QMC) takes a different route: it uses a deterministic point set that is spread evenly through axis-aligned boxes. Thus, Monte Carlo relies on independence to control variance, whereas QMC relies on geometric balance to control discrepancy.

An NC automatic sequence appears to offer a third route. It is deterministic, yet its correlation with every fixed nonzero shift tends to zero. This makes it natural to ask whether the sequence can imitate the cancellation produced by independent random signs. There is also a reason to be cautious. A d -dimensional integration rule depends on how whole blocks of coordinates are distributed, while autocorrelation measures only pairs of values along one sequence. The same NC source is therefore tested in two different roles. First, its bits are packed directly into coordinates. Second, it is used only as a digital mask for a Sobol net whose points are already well balanced.

1.2 Pattern-counting sequences

Fix a finite set A of binary words and let L be the maximum word length. Write n in binary and add $L - 1$ zeros on the left, where L is the length of the longest word in A . The symbol $\#(A, n)$ denotes the total number of overlapping appearances of words from A in this padded expansion. For an integer $k \geq 2$, define

$$a_{A,k}(n) = \omega_k^{\#(A,n)}, \quad \omega_k = e^{2\pi i/k}. \quad (2)$$

When $k = 2$, every term is either 1 or -1 . The familiar Thue–Morse and Rudin–Shapiro sequences arise from $A = \{1\}$ and $A = \{11\}$, respectively.

Here is a small example. Take $A = \{11\}$ and $k = 2$. Since the binary form of 13 is 1101, the word 11 appears once and $a_{A,2}(13) = -1$. The binary form of 15 is 1111. It contains three copies of 11 because neighboring copies are allowed to overlap, so $a_{A,2}(15) = -1$ as well. This overlap rule is important. After a new digit is read, only a short suffix can create a new pattern. That finite memory is what makes an automaton possible.

For lag $h \geq 1$ define

$$\gamma_N(h) = \frac{1}{N} \sum_{n=0}^{N-1} a(n) \overline{a(n+h)}, \quad \gamma(h) = \lim_{N \rightarrow \infty} \gamma_N(h), \quad (3)$$

whenever the limit exists. The direct-prefix program has access only to the first N generated terms, so it computes the overlap-prefix statistic

$$\hat{\gamma}_N(h) = \frac{1}{N} \sum_{n=0}^{N-h-1} a(n) \overline{a(n+h)}, \quad 1 \leq h < N. \quad (4)$$

The normalization remains N , although the sum has $N - h$ products. For each fixed lag, this convention has the same limit as (3), because

$$|\gamma_N(h) - \hat{\gamma}_N(h)| \leq \frac{h}{N} \quad (5)$$

for the unit-modulus sequences considered here. If the largest tested lag is H , the finite-prefix score is

$$G_{N,H} = \max_{1 \leq h \leq H} |\hat{\gamma}_N(h)|. \quad (6)$$

The convergence experiment uses $H = 32$. The limiting-correlation censuses use the separate and larger window $1 \leq h \leq 64$.

Definition 1.1 (Noncorrelated sequence). *A bounded sequence is noncorrelated if $\gamma(h) = 0$ for every nonzero lag h . In the finite census, the practical label NC is used when the exact recursion gives $\max_{1 \leq h \leq 64} |\gamma(h)| \leq 0.005$. This numerical rule is a classifier for the experiment; it is not a replacement for the mathematical definition.*

The quantity $\gamma_N(h)$ compares the first N terms with a copy shifted by h places. Values near 1 mean that the two copies mostly agree; values near -1 mean that they mostly have opposite signs. A value near 0 means that agreement and disagreement balance each other. For NC, this balance must hold in the limit for every fixed nonzero h . The order here matters: first choose the lag, then let N grow. A small $G_{N,H}$ on a fixed tested window is useful evidence, but by itself it cannot prove the condition for all lags.

1.3 Research questions

- Q1.** Which of the tested pattern sequences pass the operational NC test, and how can a finite-state calculation support the computed limiting correlations?
- Q2.** How quickly does a finite prefix begin to show the limiting NC behavior?
- Q3.** What happens to NC when sequences are transformed or combined?
- Q4.** Can raw NC bits be turned directly into accurate multidimensional points?
- Q5.** Do NC-based masks improve a Sobol net more than ordinary digital shifts do?

1.4 What this project contributes

The project brings together four parts of the problem:

- C1. Theory.** The report gives a finite-state description of the pattern sequences, connects their correlations with residue counts through Fourier analysis, and explains the spectral meaning of NC.
- C2. One complete proof.** Starting from the binary digit recursion, the report proves that Rudin–Shapiro is NC and also shows that the required correlation limits exist.
- C3. Reproducible experiments.** The study combines exact recursions, directly generated prefixes, spectral plots, kernel enumeration, and tests of operations on clearly stated finite collections.
- C4. A numerical-integration test.** The sequences that pass the operational NC test are used in Monte Carlo experiments. A paired design separates the effect of the mask from the effect of the underlying Sobol net.

These contributions do not all have the same logical status. The Rudin–Shapiro proof and the statement about digital shifts preserving a net are theorems. The census totals and fitted convergence rates come from finite computations. The controlled benchmark describes the selected test problems; it cannot show that one family of masks must always be best. These levels are kept separate throughout the report so that experimental evidence is not presented as proof.

1.5 Organization of the report

The report moves from the sequence itself to its possible application. Section 3 explains what NC means and how it can be proved or computed. Section 4 shows why correlation appears

in a variance formula, but also why correlation is not the same as discrepancy. Sections 6 and 7 then study the sequence experiments and the integration experiments separately. The final interpretation puts them back together. This order helps avoid confusing a one-dimensional property of a bit stream with the geometric quality of a multidimensional point set.

2 Background from the literature

2.1 Automatic sequences

An automatic sequence is produced by a finite machine that reads the base- q digits of the index. Allouche and Shallit [1] give a broad account of the algebraic, combinatorial, and number-theoretic theory. Two standard examples already show that finite-state generation does not determine correlation behavior. Thue–Morse has a singular continuous spectrum, while Rudin–Shapiro has a Lebesgue-type component and its two-point correlations vanish; see Queffelec [8]. The polynomial construction used to prove square-root cancellation goes back to Shapiro [10] and Rudin [9]. Mauduit and Sarkozy [5] explain how correlation measures can be used to judge pseudorandom behavior in finite binary sequences.

2.2 Transfer operators and digital correlation

The fixed-point calculation used later can be viewed as a finite transfer operator. Its states record dyadic residue information, and splitting the indices into even and odd parts gives the transitions. Baladi [2] develops the wider operator approach to correlation and spectral problems. Drmota [3] surveys distribution questions for automatic and digital sequences. In this project, the viewpoint is used as a computational method. The results describe the finite catalogs that were actually tested; they are not claimed as a general operator theorem for all pattern sequences.

2.3 Monte Carlo and QMC

The low-discrepancy material in this report follows Niederreiter [6]. Sobol’s construction [11] uses linear algebra over $\mathbb{F}_2$ to build digital nets that can be extended as more points are needed. Random digital shifts and scrambling keep this digital structure while producing repeated randomized estimates. The methods used here follow L’Ecuyer and Lemieux [4] and Owen [7].

2.4 Three approaches to autocorrelation

Three complementary methods are used to study autocorrelation.

1. **Finite prefixes:** generate $a(0), \dots, a(N - 1)$ and compute (3), either directly or with an FFT. This is a useful check on the implementation, although a finite prefix can only approximate a limit.
2. **Exact formulas:** some classical sequences have polynomial or substitution recursions that lead to a proof. This approach is used for Rudin–Shapiro in Section 3.
3. **Finite-state recursion:** automaticity produces a finite system for correlations conditioned on residue classes. Its fixed point gives the limiting correlations in the catalogs. Running the same system on finite dyadic blocks also gives information about the rate of convergence.

2.5 Three ways to use the sequences for integration

The integration experiments use the sequences in three ways: direct bit packing, fixed digital shifts, and randomized QMC. The last two methods keep the Sobol point set underneath the

mask; direct packing does not. This difference turns out to be essential. Autocorrelation describes pairs of values in one dimension, whereas integration error depends on the complete distribution of points in $[0, 1]^d$ and on the particular function being integrated.

3 Mathematics of noncorrelated automatic sequences

3.1 Automaticity and the 2-kernel

Definition 3.1 (2-kernel). *The 2-kernel of a sequence a is*

$$\mathcal{K}_2(a) = \{(a(2^e n + r))_{n \geq 0} : e \geq 0, 0 \leq r < 2^e\}.$$

The sequence is 2-automatic if and only if this set is finite.

Proposition 3.1. *Every sequence $a_{A,k}$ in (2) is 2-automatic.*

Proof. Read the binary digits of n from left to right. To decide whether the next digit completes a word in A , the machine only needs the previous $L-1$ digits. It must also remember the current number of detected patterns modulo k . There are at most $2^{L-1}k$ possible combinations of this information. When a new digit arrives, the machine updates the suffix and adds the number of new patterns that end there. After the last digit, the stored count determines $a_{A,k}(n)$. Thus a finite automaton is sufficient. $\square$

This proposition gives exact self-similarity, but it does not say that the output behaves randomly. Thue–Morse has a kernel of size 2 and strong correlations, while Rudin–Shapiro has a kernel of size 4 and vanishing autocorrelations. One dilation family in the experiments, called A_5 , has kernel size 28. The size of the kernel describes finite-state complexity; it does not certify NC.

3.2 Turning correlation into a residue-counting problem

Define the count difference $D_h(n) \in \mathbb{Z}/k\mathbb{Z}$ by

$$D_h(n) \equiv \#(A, n) - \#(A, n+h) \pmod{k}$$

and its empirical residue frequencies

$$p_{h,r}^{(N)} = \frac{1}{N} \# \{0 \leq n < N : D_h(n) = r\}.$$

The next identity connects the correlation sum with the residue counts used in the finite-state census.

Proposition 3.2 (Residue–Fourier identity). *For every finite prefix and lag,*

$$\gamma_N(h) = \sum_{r=0}^{k-1} p_{h,r}^{(N)} \omega_k^r. \tag{7}$$

Proof. Substituting the definition of $a_{A,k}$ gives

$$a_{A,k}(n) \overline{a_{A,k}(n+h)} = \omega_k^{\#(A,n) - \#(A,n+h)} = \omega_k^{D_h(n)}.$$

Now group the indices $0 \leq n < N$ according to the value of $D_h(n)$:

$$\frac{1}{N} \sum_{n < N} \omega_k^{D_h(n)} = \frac{1}{N} \sum_{r=0}^{k-1} \# \{n < N : D_h(n) = r\} \omega_k^r,$$

The factor multiplying ω_k^r is precisely $p_{h,r}^{(N)}$, which proves (7). $\square$

In other words, the autocorrelation is a Fourier coefficient of the histogram of $D_h(n)$. When $k = 2$, this coefficient is zero exactly when even and odd count differences occur equally often in the limit. At $k = 4$, balancing even against odd residues is not enough; the two residues inside each parity class must also balance in the required character. This is why a sequence may behave differently at different moduli.

Formula (7) turns a complex-valued cancellation problem into a finite counting problem. Rather than inspecting complex phases directly, the algorithm counts how often each possible residue of the pattern difference occurs. These residue counts are natural states for a finite automaton, so the formula becomes the basis of the census computation.

3.3 The spectral meaning of NC

Suppose that every autocorrelation limit exists. Then $h \mapsto \gamma(h)$ is a positive-definite function on $\mathbb{Z}$. Herglotz's theorem says that these values are the Fourier coefficients of a positive measure μ on the circle:

$$\gamma(h) = \int_0^{2\pi} e^{ih\theta} d\mu(\theta). \quad (8)$$

Proposition 3.3 (NC and flat spectrum). *For a unit-modulus sequence with $\gamma(0) = 1$, the sequence is NC if and only if its autocorrelation spectral measure is normalized Lebesgue measure.*

Proof. For an NC sequence, the coefficient at $h = 0$ is 1 and every other Fourier coefficient is 0. Normalized Lebesgue measure has exactly these coefficients. Since the Fourier coefficients determine a finite measure, μ must be normalized Lebesgue measure. Conversely, the nonzero Fourier coefficients of normalized Lebesgue measure vanish, so its sequence is NC. $\square$

The experiments visualize this measure through the Fejér-smoothed function

$$F_M(\theta) = \sum_{|h| \leq M} \left(1 - \frac{|h|}{M+1}\right) \gamma(h) e^{-ih\theta}. \quad (9)$$

For an NC sequence, F_M is the constant function 1. The distance of a computed profile from this flat line is summarized by its peak-to-mean ratio, its variance relative to its squared mean, and its spectral entropy.

3.4 A complete NC proof for Rudin–Shapiro

For Rudin–Shapiro, let

$$r(n) = (-1)^{\#(\{11\}, n)}.$$

The first step is to see what happens when one more binary digit is appended.

Lemma 3.1 (Digit recursion). *For every $n \geq 0$,*

$$r(2n) = r(n), \quad r(2n+1) = (-1)^n r(n). \quad (10)$$

Proof. Appending a 0 cannot create a new copy of 11, so the sign stays the same. Appending a 1 creates one new copy exactly when the previous last digit was also 1. That last digit is $n \bmod 2$, which produces the factor $(-1)^n$. $\square$

To track what the recursion does to correlations, define two finite averages:

$$\begin{aligned} \eta_N(h) &= \frac{1}{N} \sum_{n < N} r(n) r(n+h), \\ \vartheta_N(h) &= \frac{1}{N} \sum_{n < N} (-1)^n r(n) r(n+h). \end{aligned}$$

The first average is the usual correlation. The second includes $(-1)^n$ and records the parity information introduced by (10). When $N = 2M$, separate the sum over even n from the sum over odd n . Applying (10) to both parts gives the four identities below, with all quantities on the right computed at prefix length M . If N is odd, removing its last term changes either average by at most $2/N$, which vanishes as N grows. Therefore, if $\eta(h) = \lim_{N \rightarrow \infty} \eta_N(h)$ and $\vartheta(h) = \lim_{N \rightarrow \infty} \vartheta_N(h)$, the finite identities become

$$\eta(2m) = \frac{1}{2}(1 + (-1)^m)\eta(m), \quad (11)$$

$$\vartheta(2m) = \frac{1}{2}(1 - (-1)^m)\eta(m), \quad (12)$$

$$\eta(2m+1) = \frac{1}{2}((-1)^m\vartheta(m) + \vartheta(m+1)), \quad (13)$$

$$\vartheta(2m+1) = \frac{1}{2}((-1)^m\vartheta(m) - \vartheta(m+1)). \quad (14)$$

Theorem 3.1 (Rudin–Shapiro is noncorrelated). *The Rudin–Shapiro sequence satisfies $\eta(0) = 1$ and $\eta(h) = 0$ for every $h \neq 0$.*

Proof. The same recursions show both existence and vanishing of the limits. First, at lag zero,

$$\vartheta_N(0) = N^{-1} \sum_{n < N} (-1)^n \longrightarrow 0.$$

The only delicate starting point is $h = 1$. At this lag, the finite version of (14) gives the contraction

$$\vartheta_{2M}(1) = \frac{1}{2}(\vartheta_M(0) - \vartheta_M(1)).$$

Because $|\vartheta_M(0)| \leq M^{-1}$, it follows that

$$|\vartheta_N(1)| \leq \frac{1}{2}|\vartheta_{\lfloor N/2 \rfloor}(1)| + O(N^{-1}).$$

Each iteration replaces N by roughly $N/2$. After at most $\lceil \log_2 N \rceil$ steps, this gives $|\vartheta_N(1)| = O((\log N)/N) \rightarrow 0$. The corresponding finite form of (13) then shows that $\eta_N(1) \rightarrow 0$. Hence both limits exist at lag 1. The limiting identities also make the conclusion transparent: $\eta(1) = \vartheta(1)/2$ and $\vartheta(1) = -\vartheta(1)/2$, so both values must be zero.

Now use strong induction on the lag h . The induction statement includes both existence and vanishing of $\eta(h)$ and $\vartheta(h)$. For an even lag $h = 2m$, equations (11) and (12) reduce the problem to $\eta(m)$, where $m < h$. For an odd lag $h = 2m + 1 \geq 3$, equations (13) and (14) use only $\vartheta(m)$ and $\vartheta(m+1)$; both indices are again smaller than h . The induction hypothesis says that these smaller-lag limits exist and equal zero. It follows that the limits at h also exist and equal zero. Therefore every nonzero autocorrelation of Rudin–Shapiro vanishes. $\square$

Remark 3.1 (Why the proof closes). The important feature is that every recurrence moves to a smaller lag. An even lag $2m$ becomes m , and an odd lag $2m + 1 \geq 3$ becomes m and $m + 1$. Lag 1 is the only exception, so it must be handled first using the contraction above. Once that base case is established, strong induction finishes the proof. This calculation also explains the role of ϑ : ordinary correlations alone are not closed under the even–odd split, and the parity-weighted average is exactly the extra information that is needed.

3.5 Polynomial cancellation

Let $P_0 = Q_0 = 1$ and define

$$P_{j+1}(z) = P_j(z) + z^{2^j} Q_j(z),$$

$$Q_{j+1}(z) = P_j(z) - z^{2^j} Q_j(z).$$

For $|z| = 1$, the parallelogram identity gives

$$|P_j(z)|^2 + |Q_j(z)|^2 = 2^{j+1}. \quad (15)$$

Therefore $|P_j(z)| \leq \sqrt{2^{j+1}}$. The size of a sum with 2^j arbitrary terms could be proportional to 2^j , so this square-root bound demonstrates substantial cancellation in the Rudin–Shapiro polynomials. It supports the idea that the sequence has pseudorandom features. However, it does not say that points built from its bits have low discrepancy in $[0, 1]^d$.

3.6 Finite prefixes and the transfer system

For a maximum pattern length L , define the periodic correction

$$b(n) = a(n)/a(\lfloor n/2 \rfloor).$$

The value of $b(n)$ depends only on the last L binary digits of n . After conditioning correlations on suitable residue states and separating even from odd indices, one obtains a finite affine system

$$\mathbf{\Gamma} = T\mathbf{\Gamma} + \mathbf{c}. \quad (16)$$

If $I - T$ is invertible, the fixed point is $(I - T)^{-1}\mathbf{c}$, and averaging the appropriate state components gives the desired limiting correlations. Repeating the recursion on dyadic prefix lengths also shows how quickly the finite values approach that fixed point. The matrices in this project are evaluated numerically and checked against directly generated prefixes. A theorem covering every pattern set would need an exact symbolic description of all possible state systems. Numerical eigenvalues from the present catalogs are not enough for that stronger claim.

The eigenvalues nevertheless help explain the computed convergence curves. A component attached to an eigenvalue strictly inside the unit circle shrinks under repeated dyadic updates. A component attached to eigenvalue 1 can remain and produce a nonzero limiting correlation. This observation is used to interpret the calculated systems, not as a universal classification theorem.

3.7 The limits of pairwise information

NC controls every pairwise lag moment of the one-dimensional sequence. It does not determine how blocks of three or more values are distributed. For a precise comparison, let

$$\mathbf{X}_n = (a(n), a(n + h_1), \dots, a(n + h_{q-1})) \in \{-1, +1\}^q.$$

Uniform distribution of these blocks requires

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n < N} \prod_{j \in S} X_{n,j} = 0 \quad \text{for every nonempty } S \subseteq \{1, \dots, q\}.$$

NC supplies the equations with $|S| = 2$ and, when the mean is zero, the equations with $|S| = 1$. Uniform q -blocks require all nonempty choices of S , including products of three or more coordinates. NC places no general restriction on those higher-order Walsh coefficients.

A simple probability example shows the gap. Let X and Y be independent random signs and consider (X, Y, XY) . Any two coordinates are independent, but the product of all three coordinates is always 1. Thus perfect pairwise behavior can coexist with a completely rigid third-order relation. When bits with such relations are packed into several coordinates, the relation can appear as a geometric imbalance in the resulting point set. This is why NC alone cannot prove low discrepancy in several dimensions.

4 Monte Carlo and QMC background

4.1 IID Monte Carlo

Theorem 4.1 (Monte Carlo mean and variance). *Let $U_1, \dots, U_N$ be independent uniform points in $[0, 1]^d$, and suppose $\text{Var}(f(U_1)) = \sigma^2 < \infty$. Then*

$$\mathbb{E}Q_N(f) = I(f), \quad \text{Var}(Q_N(f)) = \frac{\sigma^2}{N}.$$

Proof. Each point has the uniform distribution, so the expected value of every $f(U_n)$ is $I(f)$. Linearity of expectation then gives $\mathbb{E}Q_N(f) = I(f)$. Independence makes the covariance between two different terms equal to zero. The variance of their sum is therefore $N\sigma^2$, and dividing the sum by N changes the variance to σ^2/N . $\square$

4.2 How autocorrelation enters the variance

If (X_n) is a real-valued weakly stationary process with covariance $\gamma(h)$, then

$$\text{Var}\left(\frac{1}{N} \sum_{n=0}^{N-1} X_n\right) = \frac{\gamma(0)}{N} \left[1 + 2 \sum_{h=1}^{N-1} \left(1 - \frac{h}{N}\right) \rho(h)\right], \quad (17)$$

where $\rho(h) = \gamma(h)/\gamma(0)$. The weight $1 - h/N$ has a simple counting meaning: among the first N observations, there are exactly $N - h$ pairs whose indices differ by h . Formula (17) explains why small correlations can be valuable when averaging a random process. It does not, however, make a deterministic binary stream uniformly distributed after the bits are packed into points and passed through a nonlinear function f .

To see the formula directly, expand the variance as a sum of covariances:

$$\frac{1}{N^2} \sum_{i,j < N} \text{Cov}(X_i, X_j) = \frac{1}{N^2} \left(N\gamma(0) + 2 \sum_{h=1}^{N-1} (N-h)\gamma(h) \right),$$

This is exactly (17). If every nonzero correlation is zero, the displayed second-order formula agrees with the independent case. That conclusion is limited to second-order information. A nonlinear integrand evaluated on packed blocks can depend on third-order, fourth-order, and higher joint moments that NC does not control.

4.3 Discrepancy and Koksma–Hlawka

For $P_N = \{\mathbf{u}_0, \dots, \mathbf{u}_{N-1}\}$, define

$$D_N^*(P_N) = \sup_{\mathbf{t} \in [0,1]^d} \left| \frac{1}{N} \sum_{n < N} \mathbf{1}\{\mathbf{u}_n \in [0, \mathbf{t})\} - \prod_{j=1}^d t_j \right|.$$

If f has bounded Hardy–Krause variation, then

$$|Q_N(f) - I(f)| \leq V_{HK}(f) D_N^*(P_N). \quad (18)$$

The star discrepancy in (18) is a geometric property of the whole point set, not a correlation along one sequence. It checks every axis-aligned box anchored at the origin and asks whether the observed fraction of points is close to the box volume. In several dimensions, changing one coordinate can alter membership in many such boxes. There is therefore no direct analogue of (17) that recovers discrepancy from the autocorrelation function of one bit stream.

4.4 Digital nets and shifts

A binary (t, m, s) -net consists of 2^m points arranged so that every binary elementary interval of volume 2^{t-m} contains exactly 2^t points. Sobol sequences construct these points from generating matrices over $\mathbb{F}_2$.

Theorem 4.2 (Digital shifts preserve nets). *If P is a binary (t, m, s) -net and Δ is any fixed binary mask, then the digitwise xor shift $P \oplus \Delta$ is also a (t, m, s) -net.*

Proof. A point belongs to a binary elementary interval according to a finite list of leading binary digits. Xoring every point with the same mask changes each leading-digit pattern in a one-to-one way. Thus the shift only permutes the elementary intervals of a given volume. Since no points are created or lost, every interval keeps the same point count. $\square$

Notice that the proof never uses NC. The mask may be uniformly random, derived from an NC sequence, or even derived from a strongly correlated sequence such as Thue–Morse. In every case the net parameter t is preserved. This does not mean that every mask gives the same error for a fixed integrand. A shift moves the points relative to the level sets, corners, and discontinuities of f , so the finite-sample error can still change.

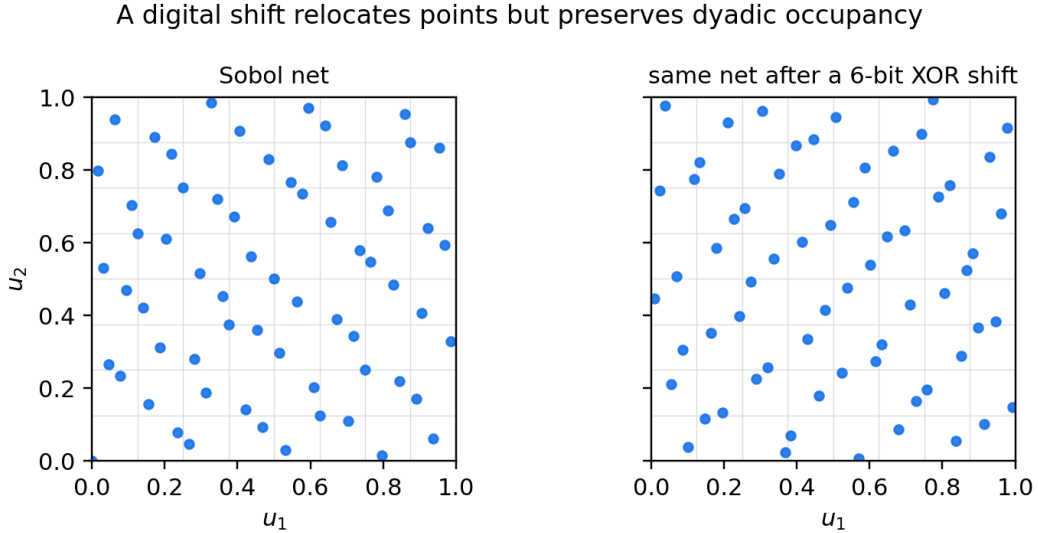


Figure 1: A Sobol net before and after a fixed xor shift. The individual points move, but every dyadic cell keeps the correct number of points. This remains true for any fixed mask, whether or not the mask comes from an NC sequence.

This observation makes the experiment harder to interpret. If Sobol plus an NC mask performs well, the improvement may come from the original Sobol net, from the general effect of shifting, or from something special about that NC mask. To study the third possibility, one must compare different masks on the same Sobol points and the same integration problems.

5 Computational methods

5.1 Two independent correlation calculations

Two independent implementations are used for the sequence calculations.

1. `recursive_exact_experiments.py` builds the finite-state correlation system and computes limiting values of $\gamma(h)$ for both named sequences and census pattern sets.

2. `experiments.py` generates the sequence terms themselves and computes finite-prefix correlations either directly or by FFT.

Because the two programs reach the answer in different ways, agreement between them is a useful check against errors in pattern counting or in the recursion. A separate runtime test fixes the convention for leading zeros.

5.2 Exact-recursion algorithm

For every pair (A, k) and every requested lag, the exact-recursion program performs the following steps:

1. compute the correction b , whose period divides 2^L ;
2. list the residue-conditioned states (h, r) that can actually be reached;
3. build the affine recursion in (16);
4. solve its fixed-point equations;
5. average the relevant residue states and record the largest score over the tested nonzero lags.

The state transitions are derived exactly, but the final linear systems are solved with floating-point numbers. For that reason, a census case is classified using the stated tolerance 0.005 on lags 1 through 64 rather than calling a small computed value symbolically equal to zero. Direct prefixes with $N \geq 16384$ give the same labels in the saved comparison file.

5.3 Building integration points

The file `nc_stream.py` turns a pattern sequence into integration points in two different ways.

Direct. Read B sequence bits for each path-coordinate cell, interpret them as an integer q , and place the coordinate at the midpoint $(q + 1/2)/2^B$ of the corresponding binary interval.

Hybrid. First generate a Sobol net. Then use the pattern sequence to build one B -bit mask for each dimension and xor those masks into every point.

The driver factory `nc_mc_driver.py` provides IID sampling, antithetic IID, Halton, Sobol, random digital shifts, the named automatic sequences, direct custom pattern drivers, and Sobol points with custom pattern shifts. Reference integrals are kept separately from the sampling code in `mc_problems.py`.

5.4 Experimental conventions and error metrics

All sample sizes are powers of two, which preserves the natural nested structure of Sobol prefixes. A convergence rate is estimated by fitting a straight line to $\log e_N$ versus $\log N$ over the listed sample sizes. Under the notation $e_N \asymp N^{-\alpha}$, the line has slope $-\alpha$. A fitted slope from only a few points can be unstable, so the exponent is never reported alone; it is paired with the error or RMSE at the largest sample size.

The reported error measures answer different questions. Absolute error describes one deterministic calculation. RMSE summarizes many runs; those runs may be genuinely randomized shifts, or they may be a fixed collection of deterministic masks when the text says so. Quantiles of absolute error show how sensitive the result is across that same collection. Only a collection drawn from a probability model justifies probabilistic error bars. Variation over a fixed catalog measures sensitivity, not statistical confidence.

5.5 Saved data and reproducibility

Every table and figure can be traced to a JSON or CSV result file. Exact correlations, census records, transformation tests, the exhaustive driver screen, and the paired benchmark are stored

separately. Keeping the outputs separate makes it harder to mix results that used different sample sizes or experimental rules.

6 Experiments on NC sequences

The sequence experiments proceed in stages. First, two finite collections are searched for NC examples. The resulting labels are then compared with the expected spectral shape and with correlations computed from long prefixes. Finally, the named sequences are transformed and combined to test how easily NC is preserved or lost. No one test answers every question. The value of the campaign comes from several independent checks pointing to the same interpretation.

Stage	Main question	Main limitation
Exact census	Which sequences in the two finite collections pass the stated NC rule?	Only the listed patterns and lags are covered
Fejér diagnostics	Do the calculated correlations produce the flat spectrum expected for NC?	This checks the calculations but is not a separate proof
Prefix convergence	How quickly do directly generated prefixes approach the calculated limit?	Only $N = 2^{10}, \dots, 2^{18}$ is tested
Transformation campaign	Which operations preserve NC in the named catalog?	The percentages depend on this particular catalog

Table 1: The sequence experiments and the limits of what each one can show.

6.1 Experiment 1: two exact-recursion censuses

The enumeration excludes words made entirely of zeros. Census A contains 846 pattern sets. First, every one- and two-element subset is taken from the 26 binary words of length at most four that contain at least one 1. Second, every three- and four-element subset is taken from the corresponding 11 words of length at most three. The total is

$$\binom{26}{1} + \binom{26}{2} + \binom{11}{3} + \binom{11}{4} = 846.$$

Census B is exhaustive within the smaller word collection: it contains every nonempty subset of those 11 words. Therefore it has $2^{11} - 1 = 2047$ sets. Every pattern set in both censuses is tested for $k = 2, \dots, 8$ and for lags 1 through 64.

	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	$k = 7$	$k = 8$
Census A, 846 sets	175	0	15	0	2	0	0
Census B, 2047 sets	640	0	66	0	12	0	2

Table 2: Numbers of census cases that pass the threshold $G \leq 0.005$. The two $k = 8$ cases appear only in the exhaustive census of words of length at most three.

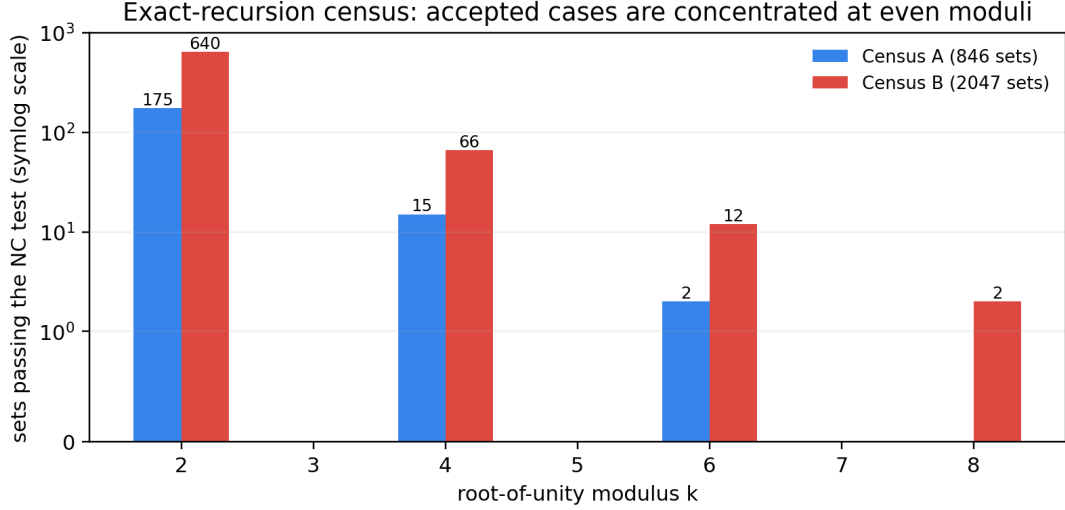


Figure 2: NC counts returned by the exact-recursion census. A logarithmic vertical scale is used so that the much smaller $k = 6$ and $k = 8$ counts remain visible.

Within these two finite searches, all accepted cases occur at even moduli. This is an observation about the enumerated pattern sets, not a proof that NC sequences cannot exist at odd moduli in a larger universe.

6.2 Experiment 2: Fejér spectra and statistical separation

The named catalog contains 26 sequences, each evaluated for moduli 2 through 8. The exact limiting correlations in (9) are used to build a Fejér profile for every sequence–modulus pair. Among these pairs, the 25 cases labeled NC have peak-to-mean ratio 1 to numerical precision. The 157 non-NC cases have mean ratio 41.84. Mann–Whitney comparisons for the peak-to-mean ratio, variance ratio, entropy, and G all give $p < 1.1 \times 10^{-15}$. On this catalog, the threshold classifier has accuracy, precision, and recall equal to 1.0.

These very small p -values show strong separation inside the constructed catalog. They should not be interpreted as population inference from independent observations, because several cases reuse the same sequence at different moduli.

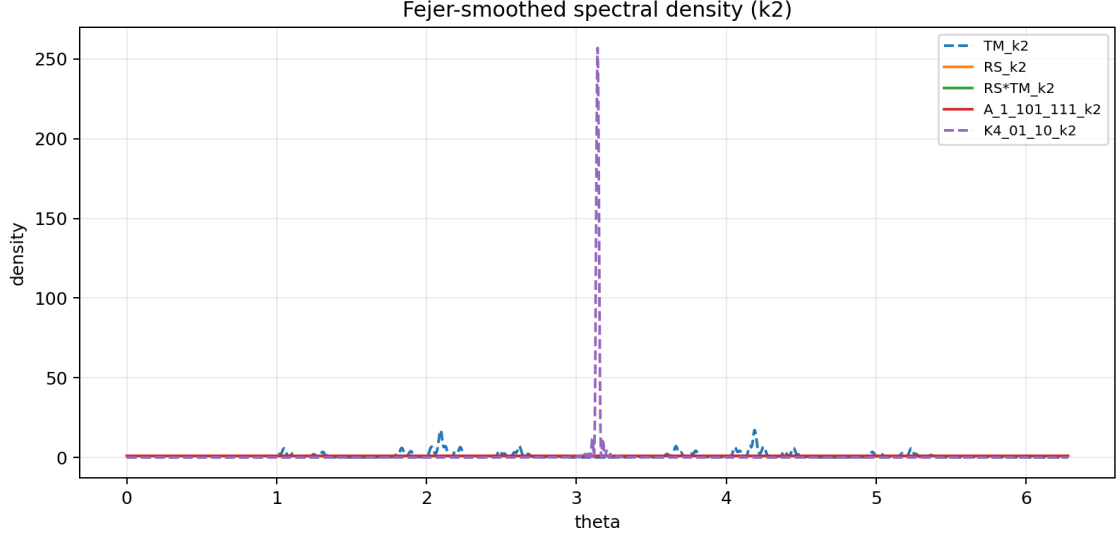


Figure 3: Fejér-smoothed profiles for $k = 2$. The examples classified as NC on the tested lag window lie on the flat line predicted by the theory. Thue–Morse and a family that is classified as NC only when $k = 4$ have visible spectral structure.

As an additional exploratory check, unsupervised K -means clustering is applied to features extracted from the Fejér profiles. Its best silhouette score is 0.749, and it finds every NC case, giving recall 1.0. Precision is low, however, because several non-NC cases are placed in the same cluster. Thus clustering may help screen candidates, but it cannot certify NC.

6.3 Experiment 3: finite-prefix convergence

Next, the terms of each named binary sequence are generated independently of the exact-recursion code. Prefix lengths range from 2^{10} to 2^{18} , and the overlap-prefix score $G_{N,32}$ uses lags 1 through 32. For the ten examples classified as NC on the limiting lag window, the fitted mean exponent of $G_{N,32}$ is 1.000. For the sixteen non-NC controls, it is only 0.0023. Equivalently, the tested overlap-prefix correlation sums for the operationally NC sequences stay bounded, so division by N gives approximately $1/N$ decay. The non-NC sums grow roughly linearly and their normalized correlations approach nonzero plateaus. This is an empirical statement about (4); it is not an additional all-lag proof of NC.

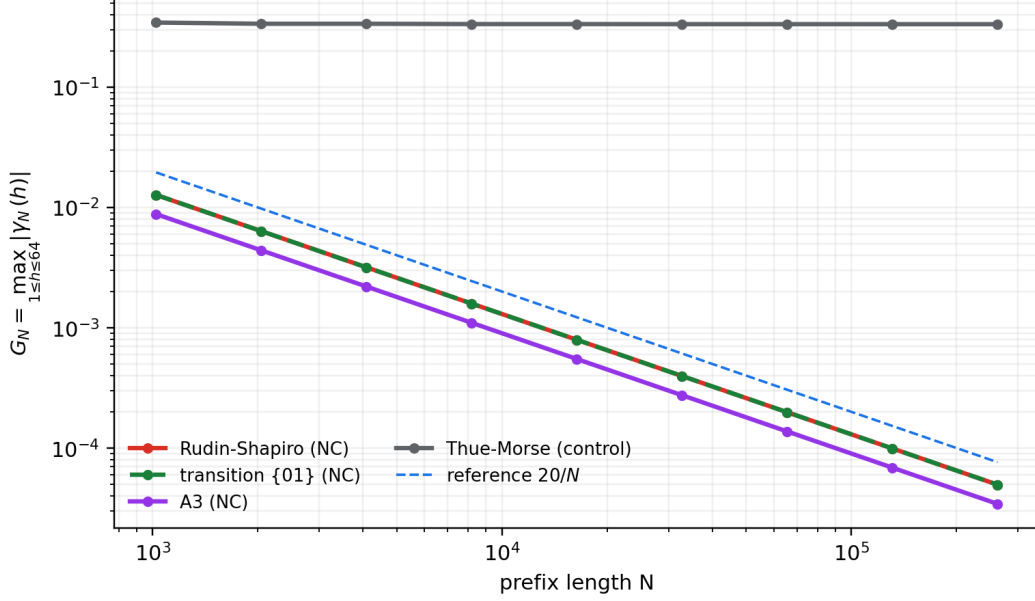


Figure 4: Overlap-prefix score $G_{N,32}$ for several operationally NC sequences and for the Thue–Morse control. The former curves follow a $1/N$ pattern over the tested prefix lengths and lags; the control approaches a nonzero level.

Class	Number	Mean α_G	Dyadic ratio	Behavior
Operationally NC binary cases	10	1.000	2.000	bounded tested sums
Non-NC controls	16	0.0023	about 1	nonzero plateau

Table 3: Summary of the finite-prefix convergence experiment on the named binary catalog.

6.4 Experiment 4: kernels and transformations

The kernel calculation gives $|\mathcal{K}_2| = 4$ for Rudin–Shapiro and 28 for the dilation example A_5 . Many non-NC families have only one or two kernel states. This suggests that state complexity may be informative, but the earlier examples show that kernel size alone cannot decide NC.

Ten operations are then applied to the sequences in the named catalog, producing 1400 result rows. When the transformed sequence is still described by a pattern set, its status is recomputed by the exact recursion. For interleavings and subsequences, prefix estimates at $N = 2^{18}$ are used instead.

Operation	NC input cases	Preserved	Closure rate
Identity, inverse, even, odd	10 each	10 each	100%
Product of distinct NC inputs	45	32	71%
Interleave distinct NC inputs	45	31	69%
Add one pattern	203	41	20%
Remove one pattern	24	2	8%
Self-product	10	0	0%
Self-interleave	10	0	0%

Table 4: Results of the transformation experiment. The percentages summarize the named catalog under the stated exact or finite-prefix calculation.

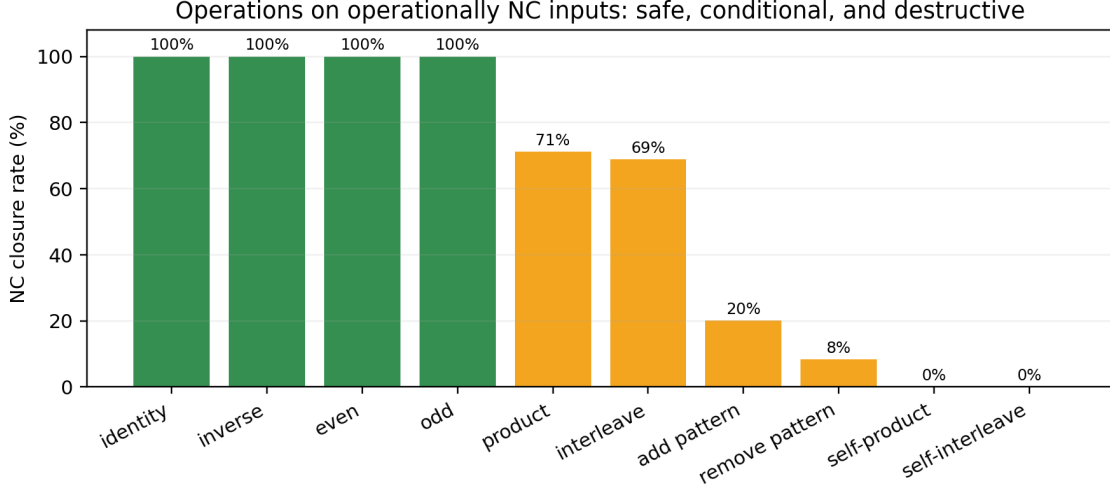


Figure 5: Observed preservation rates for the tested operations. Some operations preserve every catalog case, while others require a new NC check or destroy the property in all tested self-combinations.

The self-product always fails for an immediate reason: since $a(n)$ is a sign, $a(n)^2 \equiv 1$, and the constant sequence is fully correlated. The product of two different NC sequences is less simple. Its correlation depends on a joint Fourier coefficient of both difference processes. The relevant coefficient need not vanish merely because the corresponding coefficient of each process vanishes separately. This is another form of the higher-order problem that appears when one bit stream is reused across several integration coordinates.

6.5 Experiment 5: robustness and error controls

Three checks support the reported labels. Directly generated prefixes agree with the exact-recursion results. Correlations labeled NC produce the flat Fejér profiles predicted by the theory. Finally, the spectral statistics strongly separate the two catalog classes. The tolerance 0.005 classifies this catalog perfectly, but it is still a practical rule chosen for the finite experiment. Changing the treatment of leading zeros or increasing the lag window would create a different experiment, so those choices must always be stated.

7 Integration experiments with NC sequences

The integration study has three stages. The first asks whether the way in which the bits are used matters. The second searches for promising NC masks on several test functions. The third is the main controlled comparison: it uses equally large mask collections and the same Sobol points for every method. Each stage is motivated by a weakness or an open question from the previous one.

7.1 Test problems and sampling methods

The full code repository contains sixteen integration problems. Eight of them are selected for the focused controlled experiment and cover smooth, piecewise-smooth, kinked, path-dependent, and discontinuous integrands.

Problem	Dimension	Regularity	Reference
European call/put	1	kinked payoff	Black–Scholes closed form
Geometric Asian call	8	path dependent	lognormal closed form
$\exp(\sum x_j)$	6	smooth	$(e - 1)^6$
$\prod \sin(\pi x_j)$	6	smooth periodic	$(2/\pi)^6$
$\mathbf{1}\{\text{mean}(x) > 1/2\}$	6	discontinuous	1/2
Sobol g -function	6	piecewise smooth	1
Gaussian rare event	10	indicator	normal-tail formula

Table 5: Problems used in the focused benchmark. Additional stored tests include digital and arithmetic Asian options, polynomial and Gaussian integrals, ball and maximum indicators, expected shortfall, and quantiles.

Two different rates appear in the report:

$$\text{Type I: } G_{N,H} \sim N^{-\alpha_G} \quad (H \text{ fixed}), \quad \text{Type II: } \text{error}(Q_N) \sim N^{-\alpha_I}.$$

Type I measures a property of the source sequence: how fast its tested correlations decrease. Type II measures integration error and also depends on how the sequence is embedded, the dimension, the integrand, any randomization, and the chosen error statistic.

These rates must not be confused. A fitted value $\alpha_G \approx 1$ means that the measured prefix correlations decay quickly. It does not imply $\alpha_I \approx 1$ for a general integral. Such a conclusion would require a separate theorem controlling the distribution of the generated points. No such theorem is available for direct bit packing. In the hybrid method, the Sobol net supplies geometric control regardless of whether the mask is NC.

7.2 Experiment 6: exhaustive direct versus hybrid screen

Every driver that passes the operational NC test in the exhaustive census of patterns of length at most three is tested on a European call option with $N = 8192$. There are 720 distinct drivers. The saved output contains one direct and one Sobol-shifted result for each driver. It also repeats Rudin–Shapiro and the $k = 4$ family $\{01, 10\}$ as named direct-reference rows, so the stored direct category has 722 rows rather than 720. One IID, one plain Sobol, and one Halton baseline are recorded separately.

In the direct construction, consecutive groups of B sequence bits become binary fractions. The same stream supplies all path and coordinate cells. This is simple and reproducible, but any higher-order relation among nearby bits is transferred directly into the geometry of the points. In the hybrid construction, the sequence determines only an xor mask. The point locations before shifting, and their balance in dyadic cells, still come from the Sobol generating matrices.

Method	Count	Best error	Median error
Raw NC bit packing (includes 2 repeated references)	722	5.01×10^{-4}	2.362×10^{-1}
Sobol plus NC shift	720	1.25×10^{-6}	7.55×10^{-4}
Plain Sobol	1	4.86×10^{-4}	4.86×10^{-4}
IID	1	2.34×10^{-1}	2.34×10^{-1}
Halton	1	3.29×10^{-3}	3.29×10^{-3}

Table 6: Absolute errors in the exhaustive European-call experiment. The complete results are included among the reproducibility files.

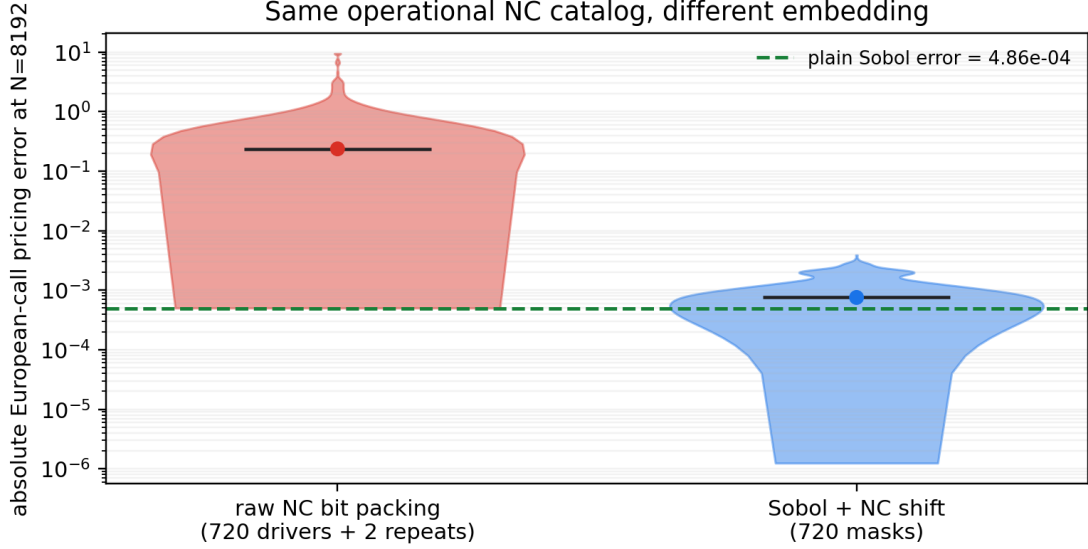


Figure 6: Errors from the same 720-driver operational catalog under two embeddings. The direct summary also contains two repeated named-reference rows. Direct bit packing is usually inaccurate, whereas using the bits only as Sobol masks gives much smaller errors.

This experiment is not a fair test of whether NC masks beat plain Sobol: there is only one plain-Sobol value and the comparison is not paired. In fact, the hybrid median is slightly larger than that one plain-Sobol error. The clear result is instead the roughly three-order-of-magnitude gap between the two ways of using the same sequences. Some individual deterministic masks also perform extremely well. The smallest error among 720 candidates was chosen after all candidates were observed, so it is a search result rather than an unbiased estimate of a typical speedup.

7.3 Experiment 7: discovery benchmark

The stored discovery benchmark uses eight problems and four powers-of-two sample sizes from 2^{10} through 2^{13} . It also uses four replicates, 16-bit coordinates, standard baselines, and eight candidate shifts derived from NC sequences. At the largest N , the best hybrid chosen separately for each problem beats the stored Sobol comparator on six of the eight problems. The mean fitted exponents are 0.694 for IID, 0.896 for Sobol, and 2.146 for the selected best method.

The last number looks impressive, but it is strongly affected by the design. Each slope uses only four sample sizes, methods use different seeds, and the winning method is selected after the errors are seen. Choosing a minimum creates optimism, and fitting a slope after that selection can increase it. For these reasons, 2.146 is neither a theorem nor an expected convergence rate for a method chosen in advance. This experiment is useful for finding hypotheses, but it is not the final basis for comparison.

Other stored experiments include ten problems with six replicates, sixteen problems at a fixed budget, forty drivers on five problems, and specialized tests for options, risk, and stochastic gradients. Their common pattern is that direct sequence-based drivers are fragile, whereas keeping a Sobol backbone is important. No automatic-sequence family wins on every problem.

7.4 Experiment 8: paired controlled benchmark

The controlled benchmark is designed to isolate any benefit that is specific to an NC mask. It makes five changes to the discovery design:

1. within a problem, every method starts from the same unscrambled Sobol prefix;

2. each automatic family and the uniform comparator contains exactly 64 masks;
3. the sample sizes are nested: $N = 512, 1024, 2048, 4096, 8192$;
4. every reference value is known analytically or from symmetry;
5. the main summary is RMSE across the full collection of shifts.

The four NC families are Rudin–Shapiro $\{11\}$, the transition sequence $\{01\}$, $A_3 = \{101, 111\}$, and $A_4 = \{1001, 1011, 1101, 1111\}$. Thue–Morse is included as a non-NC control. The uniform masks are sampled independently from the 16-bit grid. For each automatic family, the 64 masks instead come deterministically from replicate indices $0, \dots, 63$. Thus the reported “RMSE” for an automatic family is the root mean square over one fixed list. It is not an expectation under a probability law, so it does not by itself give a confidence interval.

For clarity, if the family contains $R = 64$ masks, the reported quantity is

$$\widehat{\text{RMSE}}_N = \left(\frac{1}{R} \sum_{r=1}^R (Q_{N,r}(f) - I(f))^2 \right)^{1/2}. \quad (19)$$

For uniform random shifts, this quantity estimates the RMSE under the stated random sampling rule. For an automatic family, the same formula merely summarizes the particular 64 masks indexed by $0, \dots, 63$. The numerical calculation is identical, but only the first collection is a random sample.

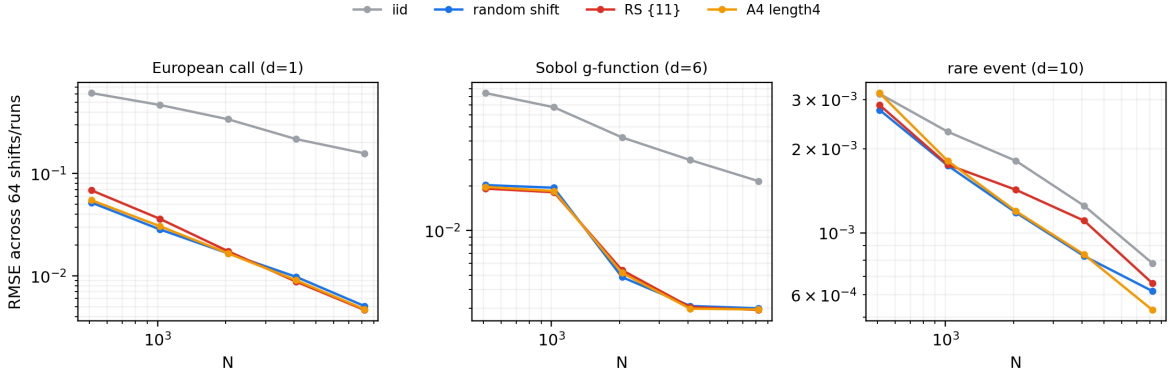


Figure 7: RMSE curves for three representative problems under the paired design. NC and uniform shifts remain close to one another, and both retain the main advantage of the underlying Sobol points over IID sampling.

Method	Mean α_{RMSE}	Median α_{RMSE}	Geom. final RMSE
IID	0.505	0.507	0.02072
Unshifted Sobol	0.849	0.891	0.001883
Uniform random shift	0.791	0.821	0.001680
Rudin–Shapiro shifts	0.809	0.850	0.001645
Transition shifts	0.835	0.906	0.001648
A_3 shifts	0.812	0.903	0.001680
A_4 shifts	0.797	0.840	0.001674
Thue–Morse control shifts	0.835	0.907	0.001749

Table 7: Results aggregated over the eight paired benchmark problems. The unshifted Sobol row contains one deterministic error curve, so it is not a replicated RMSE estimate.

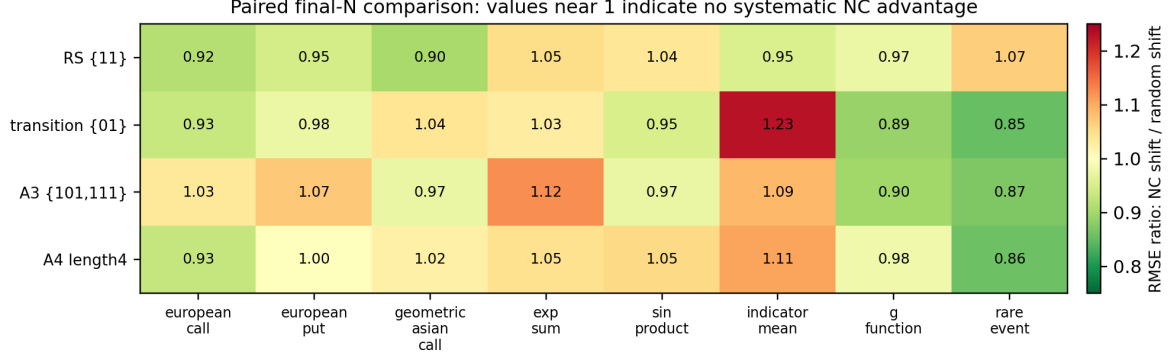


Figure 8: Ratio of each NC-family RMSE to the uniform-shift RMSE at the largest sample size. A value below one favors NC; a value above one favors the uniform random shifts.

There are 8 problems and 4 NC families, giving 32 final-size comparisons. An NC family has lower RMSE in 18 of them, or 56.25%. The geometric mean and median of the ratios are

$$\text{gmean}\left(\frac{\text{RMSE}_{\text{NC}}}{\text{RMSE}_{\text{random}}}\right) = 0.9894, \quad \text{median}\left(\frac{\text{RMSE}_{\text{NC}}}{\text{RMSE}_{\text{random}}}\right) = 0.9788.$$

Both summary values are very close to one. The fair conclusion is practical parity, not a reliable improvement caused by NC. As a check on the rate calculations, IID has fitted exponent 0.505, close to the theoretical value $1/2$.

The heatmap also shows that the outcome depends on the integrand. Some problems favor one automatic family, while others favor the uniform shifts; no row wins consistently. This agrees with the digital-shift theorem. Every mask preserves the net, but different masks move the points differently relative to the important low-order features of a particular function. A useful rule for selecting masks would therefore need information about the integrand, such as its leading Walsh or ANOVA components, rather than NC alone.

7.5 What the application results establish

Statement	What we know	Explanation
A fixed digital mask preserves the parameters of a Sobol net	Proved	The shift only permutes binary elementary intervals.
NC by itself makes directly packed points have low discrepancy	Not proved; the tested packing contradicts it	Pairwise moments do not determine block or box frequencies.
The best NC mask found in a search can beat unshifted Sobol	Observed	The winner is chosen after the errors are seen and depends on the integrand.
NC masks consistently beat uniform digital shifts	Not supported	The paired geometric-mean RMSE ratio is 0.9894, which is essentially a tie.
Random digital shifts can be used for error estimation	Standard RQMC result	They are independent randomizations of the point set.
The spread across a fixed list of NC masks is a confidence interval	False without a probability model	It measures sensitivity across a deterministic list, not sampling uncertainty.

Table 8: What is proved, what is observed in the experiments, and what the results do not support.

8 What the results mean

8.1 Why direct bit packing performs poorly

The direct method does more than read pairs of sequence values. It samples arithmetic subsequences, combines many consecutive bits into one coordinate, and uses the same automaton in several dimensions. These steps depend on higher-order relations, much like the operations that only sometimes preserve NC in the transformation experiment. The NC definition gives no control over the numbers of points falling into multidimensional boxes. The poor direct results therefore do not contradict the proof for Rudin–Shapiro; they expose a property that the proof never claimed.

The statement “raw NC streams fail” is specific to the packing rule tested here. It does not say that every possible construction from an NC sequence must be inaccurate. It says that this particular construction has no mathematical bridge from pairwise lag cancellation to the box frequencies needed for integration, and the exhaustive screen shows that the missing bridge matters in practice.

8.2 Why the Sobol hybrids perform well

Before any mask is applied, the Sobol generating matrices already give dyadic balance, useful low-dimensional projections, and nested prefixes. A digital shift moves the points without removing these properties. Consequently, Sobol points with an NC mask can work well even if NC itself adds no detectable advantage over another mask. The near tie in the paired experiment supports this explanation.

A random digital shift offers two benefits that should be separated. First, each realized mask preserves the digital net; this is true even for a fixed deterministic mask. Second, independently sampling masks creates genuinely random repeated estimates that can be used to assess uncertainty. A fixed list of NC masks provides the first benefit, but not the second unless a probability model is explicitly placed on how those masks are chosen.

8.3 Type I versus Type II

For the tested catalog, the observed relation $G_{N,32} \sim N^{-1}$ is strong evidence about the overlap-prefix correlations of the source sequences. It does not determine how quickly an integration rule converges. The experiments give examples in both directions: a Thue–Morse mask can keep QMC-like error because the Sobol net remains intact, while direct Rudin–Shapiro points can be poor because the packing rule has no discrepancy guarantee.

8.4 Answers to the research questions

- Q1.** Hundreds of pattern sets pass the NC rule in the two enumerated collections. Their limiting correlations can be computed with finite-state residue recursions. Rudin–Shapiro also has the complete direct proof given in Section 3.
- Q2.** For the named binary sequences, tested lags, and prefix lengths, the NC scores decrease approximately as $1/N$. This is a clear finite-range pattern, not a theorem about every NC sequence.
- Q3.** In the named catalog, taking the same sequence, its inverse, or its even and odd subsequences preserves NC. Reusing a sequence nonlinearly can destroy it. Products and interleavings of different sequences must be checked again rather than assumed to be NC.
- Q4.** Under the direct packing construction tested here, raw NC bits do not reliably produce accurate multidimensional integration points.

Q5. An NC mask keeps the quality guaranteed by the Sobol net. In the paired benchmark, however, NC masks do not systematically outperform an equally large collection of uniform random shifts.

8.5 Recommended practice

1. Start from a point set with a known discrepancy bound or digital-net property.
2. Use powers of two so that the nested Sobol structure is preserved.
3. When error assessment is needed, use independently randomized shifts or scramblings.
4. Treat NC masks as reproducible choices to test, not as a proof of accuracy.
5. Compare the same number of masks on the same underlying points.
6. Summarize all randomized runs with RMSE or confidence intervals; do not report only the smallest error found after searching many candidates.

9 Limitations

- **The searches are finite.** The two censuses cover 846 and 2047 pattern sets. They do not cover every possible finite set of words.
- **Only 64 lags are classified.** The practical rule stops at lag 64, while the definition of NC requires every nonzero fixed lag.
- **Some calculations use floating point.** The transition rules are exact, but the catalog eigenvalues and fixed points are evaluated numerically. Exact symbolic certificates remain a direction for future work.
- **The prefix range is finite.** The observed $1/N$ relation is fit only up to 2^{18} . It should not be presented as a universal asymptotic theorem.
- **The benchmark is limited in size.** The controlled experiment contains eight problems in dimensions 1 through 10, uses 16-bit masks, and stops at $N = 8192$.
- **The NC mask lists are deterministic.** The replicate indices choose one fixed schedule of masks. A different method of extracting masks from the same sequence may change the numerical constants.
- **Searching creates selection bias.** The best mask in an exhaustive search and the winner of a discovery benchmark are expected to look too favorable unless they are tested on new, held-out problems.
- **Not every integrand is smooth.** The indicator functions and option payoffs do not satisfy the simplest smoothness assumptions behind the Koksma–Hlawka discussion, even though QMC methods can still perform well on them.

10 Conclusions and future directions

The theory and experiments lead to six main conclusions.

1. Counting binary patterns gives a large class of automatic sequences that can be studied by finite computation. For Rudin–Shapiro, the even–odd recursion gives a complete proof that the sequence is NC.
2. The two finite censuses find hundreds of cases passing the NC rule. In the searched collections, they are concentrated at even root-of-unity moduli.
3. The ten named binary NC sequences show approximately $1/N$ decay of the tested prefix score. The non-NC controls instead approach nonzero plateaus.
4. NC can be lost when a sequence is reused algebraically. Self-products and self-interleavings fail in every tested case, while products and interleavings of different sources must be checked individually.

5. Directly packing NC bits is not a substitute for constructing a low-discrepancy point set. In the exhaustive experiment, its median error is on essentially the same scale as the IID baseline.
6. NC masks work well on Sobol points because any digital mask preserves the net. Once the comparison is properly paired, the tested NC families are essentially tied with uniform random shifts.

Future work should focus on the information that is missing between NC and multidimensional integration:

1. implement the transfer systems in exact rational or cyclotomic arithmetic and determine symbolically when their fixed points are zero;
2. measure higher-order block distributions and Walsh spectra in addition to pair correlations;
3. choose masks using information about the integrand, such as its leading Walsh or ANOVA terms;
4. randomize the initial automaton state or origin so that NC-derived masks come from a genuine probability model;
5. register the experimental rules in advance and test on a larger held-out benchmark with Owen scrambling, confidence intervals, and higher dimensions.

The conclusion is not that NC is unimportant. It is an exact and informative description of pairwise structure, and it can be computed for a wide family of automatic sequences. The Monte Carlo experiments identify the boundary of that information. To obtain an integration guarantee, one also needs control of multidimensional geometry or of higher-order relationships among the bits.

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A Reproducibility and artifact index

A.1 Core commands

```
python3 nc_one_over_n_convergence.py
python3 focused_mc_validation.py
MPLBACKEND=Agg python3 make_focused_mc_figures.py
tectonic automatic_sequences_monte_carlo_report.tex
```

A.2 Primary artifacts

Artifact	What it contains
recursive_exhaustive_nc_results.json	Exact labels for Census A
recursive_exhaustive_nc_len3_results.json	Exact labels for Census B
nc_one_over_n_convergence_results.json	Prefix scores, fitted rates, and tested transfer spectra
rigorous_fourier_all_k_results.json	Fejér features for every sequence and modulus
nc_non_nc_stat_rigor_results.json	Statistical checks and validation of the threshold
fourier_closure_operations_results.json	All 1400 transformation results
exhaustive_nc_mc_len3_results.json	European-call results for 720 NC masks
nc_mc_classical_benchmark_results.json	Results of the eight-problem discovery study
output/data/focused_mc_validation.json	Complete paired controlled benchmark
output/data/focused_mc_validation.csv	The same controlled results in a rectangular table
images/*.png	All figures included in the report

B Algorithm details

B.1 Paired digital-shift experiment

```
for problem in benchmark:
    base = one unscrambled Sobol net of size  $2^{13}$ 
    for family in [uniform, RS, transition, A3, A4, TM]:
        masks = 64 documented 16-bit masks
        for N in [512, 1024, 2048, 4096, 8192]:
            for mask in masks:
                estimate = mean(f(xor(base[:N], mask)))
            report bias, RMSE, median and 90% absolute error
```

B.2 Author verification questions

The questions below collect the main ideas that should be explained clearly in an oral presentation or defense.

1. **Why does the mixed correlation ϑ appear?** The formula for an odd index contains the factor $(-1)^n$. After the correlation sum is split into even and odd indices, ordinary correlations therefore produce a new parity-weighted average. The quantity ϑ records exactly this extra information and makes the recursion closed.
2. **Why does pairwise decorrelation not imply block uniformity?** Pairwise moments determine only first- and second-order Walsh coefficients. Uniformly distributed q -blocks require every nonempty Walsh coefficient to vanish, including products involving three or more coordinates.
3. **Why does every digital shift preserve a (t, m, s) -net?** Xoring a the same fixed mask to every point changes the leading binary patterns in a one-to-one way. It only permutes the elementary intervals, so their point counts do not change.
4. **Why is the best of 720 masks biased?** The minimum is selected after all 720 errors are known. Even equally good candidates will have one unusually small observed error by chance. A fair performance claim therefore needs a mask chosen in advance or evaluated on new problems.
5. **What distinguishes Type I from Type II?** Type I measures decay of the overlap-prefix source score $G_{N,H}$ at fixed H . Type II describes integration error, which also depends on how the bits are used, the dimension and integrand, and any randomization.
6. **Which claims are theorems?** The Rudin-Shapiro NC result and the invariance of a digital net under a fixed shift are proved. Census totals, fitted rates, transformation percentages, and benchmark comparisons are finite computational findings under the stated experimental rules.